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Auditing Grasp Feasibility Predictors with Planner-Verified Counterfactuals

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Abstract

*Grasp-feasibility predictors report near-perfect accuracy on Task and Motion Planning benchmarks, but high accuracy on these benchmarks does not verify that a predictor uses the collision geometry that the motion planner checks. We propose a planner-verified counterfactual (CF) test: holding the grasp pose fixed, we insert a solid obstacle into the approach corridor, add its points to the model input, and let a planner re-derive the label, which flips from feasible to infeasible. Across eight predictors spanning point clouds, voxels, depth, and object tokens, CF-pair AUROC falls from high natural AUROC (*0.920–0.996*) to chance (*0.470–0.504*), and the flip holds across four planners, two arms, and a real-object obstacle. Predictors fail at different resolutions, from a bounding box to a* 12.5 *cm voxel. In the point-cloud models we dissect, the obstacle either moves the representation in a direction unrelated to whether it blocks the grasp or is not registered at all. A clearance-aware read of the corridor points, trained on the unmodified benchmark with no counterfactual scenes, reaches* 0.983 *CF-pair AUROC, but it remains bound to its trained obstacle placements; a placement-invariant clearance representation is still open. We release the counterfactual labels, metrics, datasheet, and relabeling pipeline at* https://github.com/codewithsajid/graspfc-cf.

1. Introduction

Computing grasp poses (grasps) is a fundamental problem in robotic manipulation. Given a grasp on an object, the end-effector can reach it only if a collision-free path exists from the robot's start configuration; motion planning is *feasible* when such a path exists and *infeasible* otherwise. Sampling-based planners [18] commonly used in manipulation are only probabilistically complete, so they cannot certify infeasibility: they keep searching until termination or timeout, inflating overall computation time.

Learned predictors [1, 2, 5, 34, 35] that score this *feasibility* from a scene are now a standard speed-up for Task and Motion Planning (TAMP) [6, 13, 17, 29], and on point-cloud benchmarks they report near-perfect accuracy.[1] However, these benchmarks evaluate predictor models on naturally sampled grasps, for which the grasp pose is often sufficient to predict the label. A predictor can therefore saturate the benchmark without using the full scene geometry checked by the motion planner. Fig. 1 shows one such grasp: the initial feasible grasp (left) is made infeasible by inserting the cube (right), yet the predictor's feasibility score remains unchanged.

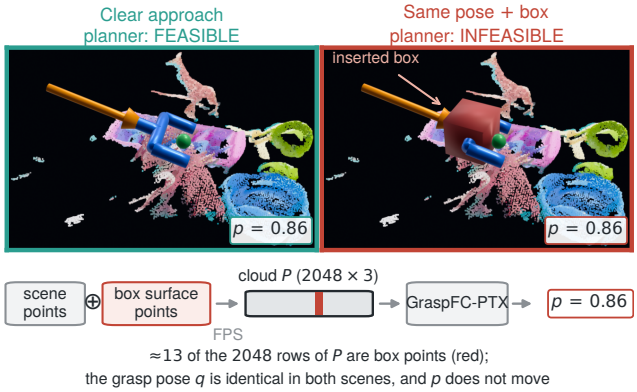


Figure 1. The counterfactual test. (**left**) A feasible grasp. (**right**) The same grasp becomes infeasible once a cube (red) is inserted in its approach path; points sampled from the cube's surface are added to the point-cloud input. The classifier (GraspFC-PTX) still predicts $p(\text{feasible}) = 0.86$ in both scenes. Over 1,065 such pairs, its CF-pair AUROC is 0.498 (chance = 0.5).

Collisions are a frequent and significant source of grasp infeasibility (Sec. 3). A natural question is *why standard benchmarks fail to reveal collision.* Their sampling rarely includes examples near the boundary where a small change in geometry causes a grasp to switch from feasible to infeasible (Sec. 4.1). Consequently, high accuracy on the benchmark does not verify that the model reasons about collisions. The model can achieve high accuracy without actually learning this aspect of grasp feasibility. This exposes

[1] TAMP combines discrete actions of task planning with the continuous geometric motion planning. A task-level action may turn out to be infeasible at the motion-level, e.g., objects obstructing a grasp, requiring task-level re-planning.

two related issues: shortcut learning [16] and causal confusion [7], both of which can be measured directly through controlled interventions.

We evaluate different models using their released weights on their original benchmarks and robots. The models do not fail in the same way. Instead, each responds only at the level of detail its supervision, query formulation, and representation support, and stops below it (Sec. 4.3, Tab. 4). Importantly, the obstacle is present in the input representations. Thus, this insensitivity cannot simply be attributed to PointNet's critical-point-set theorem [25], which concerns the learned per-point representation rather than the geometric structure of the scene. We verify this directly in Sec. 4.4. Further evidence comes from a clearance-supervised model trained on the same unmodified benchmark, with no counterfactual data at all, which still recovers the collision term (Sec. 4.5).

Together, these results show that the relevant collision signal is present in the input, but standard supervision does not require the model to use it.

Contributions. We audit eight predictors: three classifiers of a grasp-feasibility benchmark [3] built on GraspNet-1B scenes [11], and five prior methods spanning point clouds, voxels, depth images, and object tokens [1, 2, 5, 34, 35]. For [2, 3, 5, 35], we additionally evaluate the authors' released weights on their original data and robot platforms. *These results leave open a fundamental question: do the predictors actually capture collision-relevant scene geometry?* We introduce a systematic test, along with metrics and data, to measure this correspondence.

- **Benchmark saturation and collision limits.** Predictors that are near-perfect on the natural benchmark score 0.47-0.504 once we insert an obstacle that flips the collision label at the same grasp (Sec. 4.1, Tab. 1). Natural data almost never places two grasps at the same pose on opposite sides of a contact boundary, so a predictor can score well without reading the scene (Sec. 4.1).
- **Planner-verified counterfactual test and metrics.** We define the *corridor insertion* edit and verify both of its preconditions: every edit flips the planner's label, and the inserted obstacle survives the model's downsampling (Sec. 3). We report CF-pair and within-grasp AUROC, and release pick-clear accuracy for auditing new predictors. The planner-swap and real-object controls show the flip is not an artifact of one planner, arm, or obstacle type (Sec. 4.2).
- **Predictors fall short in different ways.** Each responds only down to the resolution its representation and supervision support, from a bounding box to a 12.5 cm voxel, with two point-cloud predictors not resolving the obstacle at all (Sec. 4.3, Tab. 4). Three do so on their authors' own data and robot. We therefore report each at its own resolution rather than as one aggregate score.
- **Collision signal in the input and supervised read.** A clearance-supervised read of the same points reaches CF-pair 0.983, trained on the unmodified benchmark and never shown a counterfactual scene (Sec. 4.5), consistent with ClearanceNet [20].
- **A released counterfactual benchmark.** Each label costs a planner call, so we release the GraspNet-1B counterfactual set with planner-verified labels, a datasheet [14], and the relabeling pipeline,[2] with each obstacle family as its own split. A new predictor can be scored on it without running a planner (supp. Appendix I).

2. Related Work

Learned grasp and motion feasibility. Predicting whether a grasp is reachable and collision-free is widely used to prune candidates before planning, where collision checking is the dominant runtime cost [36]. GraspNet-1B [11] provides real cluttered scenes and grasp annotations, atop a mature grasp-synthesis literature [24, 28]; a recent benchmark labels its grasps for motion feasibility with a planner [3]. Feasibility and collision predictors span point clouds, voxels, depth, and object tokens [1, 2, 5, 20, 23, 34, 35]. SceneCollisionNet [5] learns an implicit collision function for rearrangement, queried on a voxelized scene. PIGINet [35] scores the feasibility of a whole task-level plan from object tokens, and is never queried on a single approach corridor. NFC [34] and AGFP-Net [1] classify feasibility from depth for multi-robot manipulation planning. GRN [2] reasons over per-object bounding boxes and has an explicit obstruction head. Each reports high accuracy on its own benchmark, which we do not dispute. *We ask a different question: does the benchmark test whether a predictor uses the collision geometry the motion planner checks?*

Benchmark auditing, shortcut learning, and counterfactuals. Models that reach high accuracy through a spurious correlate rather than the intended signal, lead to recurring failure modes [15, 16]; additional inputs can make it worse, once it latches onto a non-causal cue [7]. Uninformed baselines and stress tests such as HANS [22] expose these shortcuts. Contrast sets [12] and counterfactually-augmented data [19] edit input minimally, then re-derive its label. Similar failures recur in 3D vision and robot learning, e.g., point-cloud networks latching onto rotation-frame cues [8]. Generalist policies generalize poorly when the bottleneck is dataset diversity rather than scale [9, 33]. Our test shows that an uninformed baseline model *can* succeed without the input and therefore may lead to *wrong* predictions when the input alone is changed; a motion planner, not

[2]Repository (code, data, and datasheet): https://github.com/codewithsajid/graspfc-cf.

a human annotator, certifies the new label.

Clearance as a supervised quantity. ClearanceNet [20] learns the clearance between a robot configuration and the scene, and uses the learned estimate to accelerate batched motion planning. Its lesson is that collision geometry is learnable once the query and the label are both stated in terms of clearance. A feasibility benchmark does not impose that condition. Its binary label supervises the outcome, not the geometric margin that produces it. We show the other half of that picture. Nothing in the benchmark forces a predictor to use the collision term, because a pose shortcut fits its labels just as well. The labels are still sufficient: supervising clearance over the same points, on the benchmark's own splits, recovers the term (Sec. 4.5).

3D encoders and tokenizers. Point-cloud recognition is dominated by permutation-invariant encoders with max pooling [25] and their hierarchical [26, 27] and transformer [30] successors. PointNet's critical-point-set theorem [25] is often read as making such encoders insensitive to added points. The condition is on the learned per-point map rather than on geometry, and it holds only for a single global max-pool with no local or positional branch. We measure it directly and find that the inserted points do move the pooled feature (Sec. 4.4). Self-supervised pretraining by masked reconstruction [37] or contrastive objectives [31] learns recognition features, which need not encode a contact boundary. Reconstruction tokenizers are the one case where the obstacle is lost outright (Sec. 4.4).

3. Counterfactual Methodology

The test requires four components: a fixed grasp, an insertable obstacle in the approach corridor, a planner to recompute the label, and a model input incorporating the obstacle points. With these in place, a single scene edit isolates the collision term while leaving all other factors unchanged. We evaluate on a shared benchmark and within each audited method's native environment and robot (Sec. 4.3).

The shared benchmark and its oracle. The feasibility benchmark of [3] is built on GraspNet-1B [11]. It has 190 real RGB-D tabletop scenes with official Seen, Similar, and Novel object splits. Each grasp carries a label from an RRT-Connect planner [21] oracle for a 7-DOF Franka Panda in PyBullet [4]. A grasp is *feasible* only when the planner finds a collision-free path to the pre-grasp pose (10 cm from grasp pose) from its home configuration. This gives 2,706,160 per-grasp labels, 9.6% of them feasible. The label is effectively deterministic: re-running the planner with 8 seeds on held-out grasps gives $0/480$ flips, so label noise is at most 0.6% at 95% confidence. The oracle is not the variable either. Labels do differ across planners and arms, so we vary both: four planning algorithms and two kinematic chains, the 7-DoF Panda and a 6-DoF xArm6. We call this the *planner-swap control* (Sec. 4.2, Tab. 3).

Collision decides most infeasibility. Across all labeled grasps, collision accounts for 64.3% of infeasibility: 49.0% from blocked transit and 15.3% from blocked approach. The remaining 35.7% is due to reachability. We assign each infeasible grasp to the first binding constraint in pipeline order, making these collision rates conservative lower bounds.

The edit. A perturbation is useful only if it changes the label and leaves a physically valid scene. We hold the grasp pose fixed, drop a solid box into the 10 cm approach corridor, re-run the planner for the true new label, and add the box points to the model input (Fig. 2). We call this edit a *corridor insertion*. The box has edge 1.5 to 5 cm and sits at a 5 to 15 cm standoff. Two properties are verified. First, the edit causes a real flip: it relabels every pair, while a matched box placed 30 cm off-corridor leaves the grasp feasible 98.0% of the time. Second, the obstacle survives downsampling, since every item retains at least one box point, ≈ 13 on average, after farthest-point sampling to the 2048-point input. A graded variant slides the box through six corridor offsets, from blocking to harmless, so a model must track *where* the obstacle is.

Obstacle families. The corridor box is the primary family, but we also vary the geometry, including spheres, non-convex L-shapes, table-mounted pillars, transit-path boxes, and four obstacle clusters. Size, standoff, and placement are independently swept, spanning six corridor offsets from blocking to harmless, plus adjacent, distal, and on-axis placements. A no-primitive family, the *real-object control*, uses a real benchmark object placed beside the target. Each family is planner-verified and released as a separate audit split (Sec. 4.2, Sec. 4.5, supp. Appendix I); geometry and the sampling details are specified in supp. Appendix H.

Metrics. We report *natural AUROC* on the original splits without inserted obstacles. *CF-pair AUROC* ranks clear versus blocked versions of the same grasp at an identical pose, while *within-grasp AUROC* ranks feasible versus infeasible obstacle offsets. Chance is 0.5 for both. The audit suite additionally reports *pick-clear* accuracy: among two candidates where one is blocked, how often the clear grasp is ranked first (supp. Appendix I). Pose-only models cannot exceed chance on any metric, since paired examples share the same pose. CF-pair AUROC uses 1,065 released pairs, with bootstrap 95% CIs over pairs; trained results use 3 seeds. Thus, scores near 0.5 reflect genuine failure to detect collision, not estimation noise.

Models and evidence regimes. We evaluate three model groups (Tab. 1). **(i)** The three GraspFC classifiers of [3], run from their released weights on the shared benchmark. These are GraspFC-NNet, -Conv3D, and -PTX, the last on a PTv3 [30] backbone. **(ii)** Four prior representations, re-implemented and trained on the planner labels under one protocol: SceneCollisionNet, PIGINet, NFC [34], and AGFP-Net [1]. They share the oracle, the splits, and the

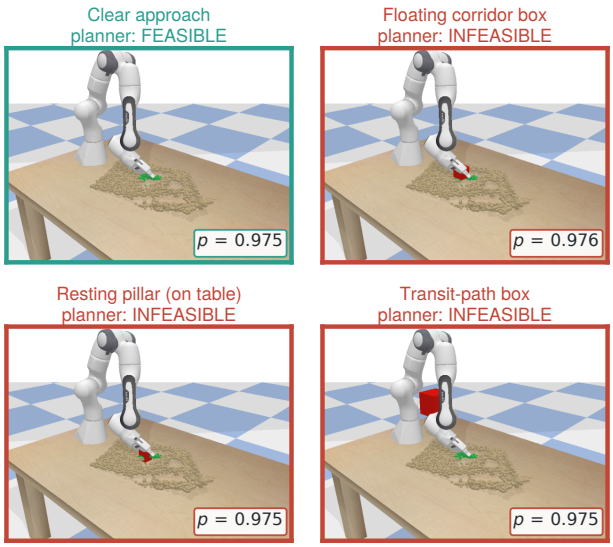


Figure 2. The counterfactual protocol on a GraspNet-1B scene (deployed GraspFC-PTX). One grasp is held fixed; each of three obstacles (a floating corridor box, a resting pillar, a transit-path box) makes the planner relabel it infeasible (red), while the clear grasp stays feasible (green). The model returns the same p(feasible), 0.975 to 0.976, for all four, so its score does not separate the clear scene from any of the three blocked ones.

loss, so this group tests the representation rather than the authors' deployed system. **(iii)** Released weights of published methods run on their own data (Sec. 4.3). Per-model architectures and losses are in supp. Appendix C. Every label on the shared benchmark comes from the same RRT-Connect oracle, which we verify deterministic and causal (supp. Appendix F).

4. Experiments

4.1. Benchmark saturation and collision sensitivity

Tab. 1 reports the audited predictors. All are near-perfect on the natural benchmark, but this score does not distinguish predictors that use collision geometry from those that do not. On same-pose counterfactuals, every predictor scores 0.47-0.504 AUROC (within-grasp 0.47 to 0.52). GRN is the one predictor whose score changes at all, and only to obstacle presence: it cannot resolve the sub-bounding-box shape that determines collision (Fig. 6), so its entry (*) is a shape swap rather than an insertion. This failure is not due to input modality. The obstacle appears in the voxel grid of GraspFC-Conv3D and in all five depth views of AGFP-Net, yet both discard it; NFC uses only the grasp pose and never its neighbor channel (Tab. 4). The deciding geometry is therefore present in the inputs and can be recovered from the same points (Sec. 4.5).

Why natural labels do not require collision. Feasibility labels are geometrically separated, with labels flipping at contact boundaries that natural sampling rarely straddles, making same-pose contrast essentially absent. Conditioning on grasp difficulty, which carries no scene information, therefore removes most of the model's apparent advantage. This matches the benchmark's own observation that grasp pose rather than scene geometry carries the feasibility signal [3]. The *input* holds the collision geometry, as shown in Sec. 4.5, recovering it under the benchmark's own labels. The *label distribution* never requires a predictor to use it, so no quantity of natural data forces the term into the model. This follows from how the labels distribute, not from one benchmark's design.

Predictor	Input	Par.	Regime	Nat.	CF-pair
GraspFC-NNet [3]	pose+crop	1.65M	released	0.995	0.491
GraspFC-Conv3D [3]	pose+voxels	9.7M	released	0.976	0.497
GraspFC-PTX [3]	pose+cloud	9.0M	released	**0.996**	0.498
SceneCollisionNet [5]	voxels	26M	re-impl.+rel.	0.964	0.470
PIGINet [35]	tokens	31M	re-impl.+rel.	0.960	0.500
NFC† [34]	depth	22.8k	re-impl.	0.965	0.504
AGFP-Net† [1]	depth views	33M	re-impl.	0.920	0.498
GRN [2]	box+pose	2.3M	released	0.995	0.500*

Table 1. The eight audited predictors: input modality, parameters, evidence regime, and natural vs. counterfactual AUROC on the shared benchmark (GRN on its own, below the rule). *released*: the authors' released weights; *re-impl.*: re-implemented and trained on the planner labels ("+rel.": also run from released weights on its authors' own data, Tab. 4). †No public code or weights; re-implemented from the published description. *Nat.*: ranking feasible vs. infeasible grasps on the unmodified benchmark. *CF-pair*: ranking the clear vs. obstacle-inserted version of a grasp at the identical pose, on the planner-verified pair set (n=1,065); 0.5 is the no-signal baseline. *GRN detects presence but does not resolve sub-bounding-box shape, so its entry is the same-bounding-box shape swap (Tab. 4). The methods' original (non-AUROC) headlines and full interventional scores are in supp. Tab. 9 and Appendix D.

4.2. Robustness to placement, obstacle, planner, and robot

The result is not an artifact of our re-implementations or of one obstacle location. Re-scoring *inference-only*, with no model seeing a new label, we move the obstacle to held-out scenes and unseen placements. These are an adjacent in-crop offset (CF-adjacent, $n = 561$) and a distal out-of-crop offset (CF-distal, $n = 250$). Every model stays in $[0.47, 0.554]$ at every placement, and its score barely moves with or without the obstacle (Tab. 2). Fig. 3 shows the dual fact for the *cured* models, those we fine-tune or train on planner-relabeled counterfactual pairs. Counterfactual training makes them react only within a shell around the gripper. No model we test, cured or not, holds a clearance representation that survives moving the obstacle across the

Model (run as-is)	CF-iid	CF-adj.	CF-dist.	max $\lvert\Delta p\rvert$
GraspFC-PTX (9.0M)	0.501	0.499	0.497	<0.001
PIGINet	0.500	0.508	0.493	0.014
NFC	0.504	0.511	0.532	0.014
AGFP-Net	0.498	0.497	0.499	0.007
SceneCollisionNet (26M)	0.470	0.473	0.554	0.052

Table 2. Placement-invariance (inference-only, no retraining). CF-pair AUROC with the obstacle at the in-distribution corridor offset (iid), an unseen adjacent offset (n=561), and a distal offset outside the crop (n=250), all in held-out scenes. The one cell above 0.5 at distance (SceneCollisionNet, 0.554) is ≈ 1.4 SE from 0.5 and falls below it at the other two offsets. max $\lvert\Delta p\rvert$ is the largest score change the obstacle produces.

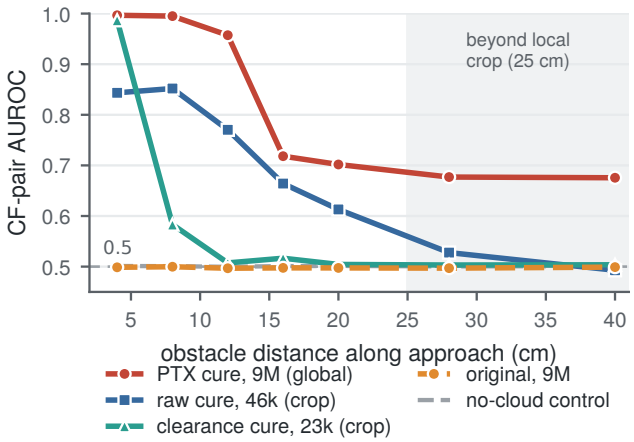


Figure 3. CF-pair AUROC vs. obstacle distance along the approach (inference-only). The three counterfactual-cured variants (the deployed global PTX, and two crop-input models using raw corridor points vs. clearance-aware tokens) react only near the gripper; the original model and a no-cloud control stay at 0.5.

workspace. The result also holds across standoff and obstacle type, shown below, and across dense multi-object clutter (supp. Fig. 12). Most tellingly, it holds in the real-object control, where the obstacle is not a foreign primitive at all: a GraspNet-1B object from the same Novel split, resting on the table next to the target, flips every one of 150 grasps, while a matched copy set aside leaves all of them feasible. The deployed model returns essentially the same score either way ($\lvert\Delta p\rvert = 0.001$; Tab. 3, supp. Fig. 10). The diagnosis is therefore not an out-of-distribution obstacle artifact.

Oracle, obstacle type, and embodiment. We re-run the planner-verified counterfactual under three standoffs and two further obstacle types, and with an in-distribution dataset object from the test splits placed in the grasp approach direction (supp. Fig. 10). The deployed model is re-scored on each (Tab. 3). The *resting* case is gravity-plausible: a pillar standing on the table that rises into the

Oracle / obstacle	Flip	Null	CF-pair	$\lvert\Delta p\rvert$
Corridor box, 10 cm standoff	1.00	0.02	0.498	0.001
Corridor box, 5 cm standoff	1.00	0.02	0.499	0.000
Corridor box, 15 cm standoff	1.00	0.02	0.500	0.000
Resting pillar (on the table)	0.74	0.03	0.499	0.000
Transit-path box	0.56	0.01	0.491	0.001
Dataset object (on the table)	1.00	0.00	0.500	0.001
Planner swap (4 planners × 2 arms, 8,412 planner calls): flip 1.000 in 8 of 8 conditions; label agreement 0.973–0.995; Cohen's κ 0.940–0.990.				

Table 3. Robustness of the flip and of the model's insensitivity to oracle standoff, obstacle type, planner, and arm. *Flip*: fraction of grasps the inserted obstacle relabels infeasible (planner ground truth). *Null*: false-flip rate of a matched off-corridor box that should stay feasible (ideal 0). *CF-pair* (chance 0.5) and $\lvert\Delta p\rvert$ (score change with vs. without the obstacle) are for the deployed model (GraspFC-PTX). The planner relabels every variant while the model stays at chance. Labels themselves differ across planners and arms (607 of 957 Panda-feasible grasps are out of xArm6 reach); the flip does not. Per-condition values: supp. Tab. 7.

descent corridor, not a floating body. The third case lies in the arm's *transit* path, the dominant 49.0% blocked-transit failure mode. The edits therefore span both feasibility terms and are physically realizable. In every variant the planner relabels the grasp, with causal flip 0.56 to 1.00, while a matched off-corridor null box leaves it feasible $\geq 97\%$ of the time. The deployed model stays near chance throughout (CF-pair AUROC 0.491 to 0.500) and returns an essentially identical score with and without the obstacle ($\lvert\Delta p\rvert \leq 0.001$). Dense clutter behaves the same: a cluster of four obstacles flips all 150 re-checked grasps, and the model stays at chance (CF-pair 0.498, $\lvert\Delta p\rvert < 0.001$; supp. Fig. 12).

The oracle itself is not the variable either. The planner-swap control relabels the grasp in 8 of 8 conditions at flip rate 1.000 (Tab. 3). The diagnosis therefore tracks the representation, not the standoff, the obstacle shape, the planning algorithm, or the arm.

4.3. Released weights on the authors' own data

Most directly, we run published models from their *own released weights on their own benchmarks* (Tab. 4 and Figs. 4 to 6), which removes the re-implementation caveat. The verdicts are not uniform. Each model responds at the resolution its supervision, native query semantics, and representation provide, and stops responding below it (Tab. 4). PIGINet, on its authors' PR2 kitchen, calls a planner-blocked pick feasible 90% of the time against a ground truth of 0% (Fig. 4; aggregated in supp. Fig. 14). It is a sequence-level predictor over object tokens and was never queried on an approach corridor, and the shortcut extends to multi-

Predictor	Responds to	Stops below	Native	Interv.
GRN†	presence, bbox	sub-bbox shape	0.995	0.500
SceneCollisionNet†	contact, coarsely	12.5 cm voxel	0.969	**0.821**
PIGINet†	object tokens	corridor geometry	0.86	0.470
GraspFC-NNet	presence	whether it blocks	—	0.491
GraspFC-Conv3D	voxel grid	any obstacle	—	0.497
GraspFC-PTX	(no response)	any obstacle	—	0.498
NFC	grasp pose only	its neighbor channel	—	0.504
AGFP-Net	5-view depth	any obstacle	—	0.498
Clearance tokens (ours)	corridor clearance	off-family placement	—	**0.983**

Table 4. The physical-sensitivity spectrum on one protocol, for all eight audited predictors. *Responds to* is the coarsest geometry that moves the score; *stops below* is where the response no longer tracks the planner. †Released weights on the method's own data and robot: *Native* is our AUROC, *Interv.* the counterfactual score on that same set (GRN and SceneCollisionNet: shape swap; PIGINet: insert-a-box). Each reproduces its authors' published headline first, none of which is an AUROC: GRN's action, grasp, and IK F1 and obstruction MAE [2], SceneCollisionNet's trajectory accuracy and AP [5], PIGINet's validation plan-feasibility and planning-time cut [35]; values in supp. Tab. 9. The remaining rows are on the shared benchmark, where *Native* is Tab. 1's Nat. column, so *Interv.* compares within a regime and not across. GraspFC-NNet and GraspFC-PTX differ by 28× in how strongly the obstacle moves their score ($|\Delta p|$ = 0.027 / 0.001), yet neither tracks collision. Per-model mechanisms are in supp. Tab. 10.

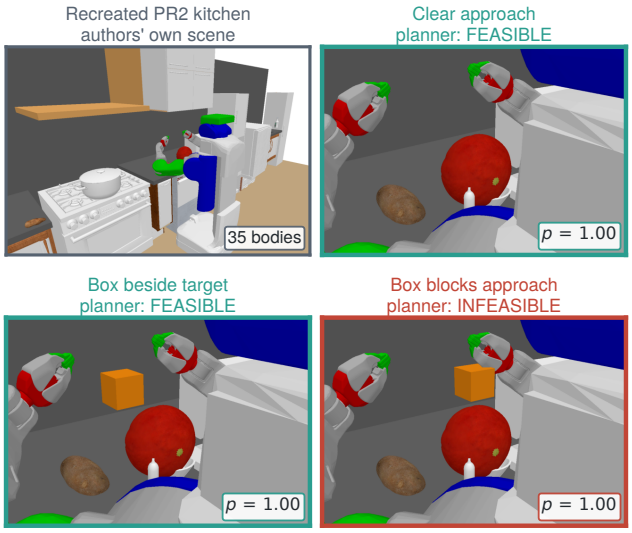


Figure 4. PIGINet on the authors' own PR2 kitchen (35-body saved scene, top-left; released weights). At one fixed arm pose: a clear approach, a harmless box beside the target, and a box blocking the approach. The planner relabels only the blocking case infeasible (red), yet PIGINet returns the identical $p = 1.00$ in all three, so its score does not register the corridor box.

step plans (supp. Appendix A). SceneCollisionNet, on its own ACRONYM [10] scenes, is the one audited predictor that responds to shape above chance (0.821 on a same-bounding-box swap). It is not a decision boundary. Swept from a deep overlap to a 6 cm clear gap, its score holds at $p_{\text{coll}} = 0.76$ straight across the true contact boundary (Fig. 5). It reads contact, and stops at its 12.5 cm voxel (supp. Appendix B). GRN reads a bounding box by design. It *detects* inserted geometry, yet scores a box and its inscribed same-bounding-box ellipsoid identically: a *shape-swap AUROC* of 0.500, with two further geometry holes in supp. Appendix E. One aggregate number over these predictors would hide five mechanisms with five different remedies.

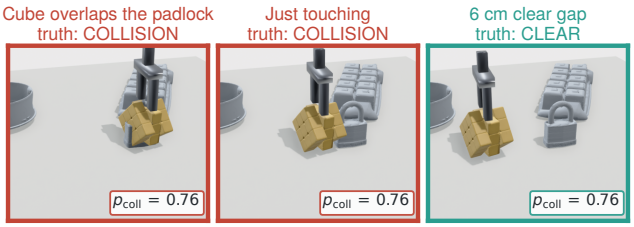


Figure 5. SceneCollisionNet on a cluttered ACRONYM-style tabletop, released checkpoint. A gripper-held cube is swept against a solid obstacle from overlapping it, through contact, to a 6 cm clear gap. The true label flips (red to teal), yet the score holds at $p_{\text{coll}} = 0.76$ (0.75 to 0.77 over the sweep, never below 0.5), where an ideal checker steps from 1 to 0. The whole sweep is shorter than the 12.5 cm voxel it reasons over.

4.4. Where the collision term is lost

If the deciding evidence is in the input, where is it lost? Not at the pooling step, and not by construction. PointNet's critical-point-set theorem [25] does not say that points inside a critical set's hull cannot change the global feature. It gives a critical set and an upper-bound set, and adding points leaves the pooled feature unchanged exactly when no added point raises a pooled channel. That is a condition on the learned per-point map, not on geometry. We measure it on GraspFC-NNet's max-pooled branch in isolation, over the planner-verified pairs, and the condition fails. Only 16% of the inserted points raise no channel, and the obstacle raises 31.5 of 256 channels. The pooled feature moves *more* than a mean pool would move it (0.174 against 0.099), because the inserted points are extremal.

The obstacle therefore enters the representation. Decomposing the deployed forward pass shows how far it gets. Inside the full model that same max-pooled branch moves by 0.074 and the scene branch by 0.212. The head is not frozen either, with a per-pair $|\Delta\text{logit}|$ of 0.365. What the movement lacks is sign. The mean Δlogit is only $+0.017$, and the alignment between the feature displacement and the

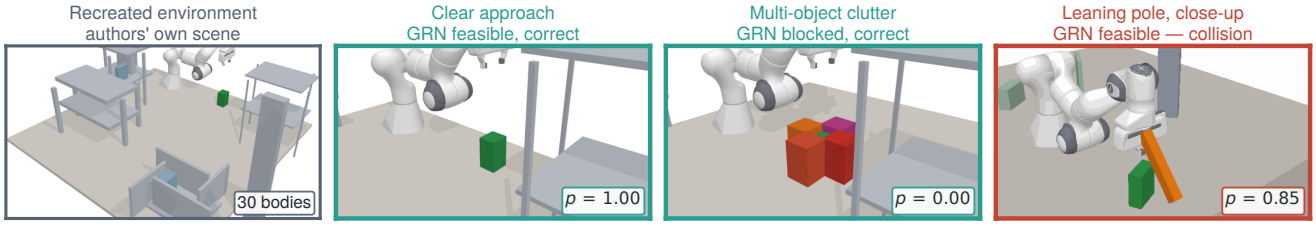


Figure 6. GRN on its own panda_3d_4 scenes (30-body environment, left; released weights). GRN detects inserted geometry, dropping from feasible 1.00 (clear) to 0.00 when a multi-object cluster blocks the approach. What it does not resolve is sub-bounding-box shape (rightmost, close-up from a rollout): a thin pole leaning 40° into the approach corridor is read as an upright box and called feasible ($p = 0.85$), yet the gripper drives straight through it. GRN reads the bounding box, not the shape (Tab. 4).

one direction the head can read has mean cosine $+0.043$ against a magnitude of 0.251. The response is large and direction-random. We test this directly by placing two obstacles at the identical grasp pose, adding the same number of points to the same cloud, one blocking the approach and one 30 cm off-corridor. GraspFC-NNet registers both strongly ($|\Delta p|$ 0.027) and GraspFC-PTX registers neither (0.001). Neither ranks the blocking one below the harmless one (blocking AUROC 0.481 and 0.502, bootstrap intervals containing 0.5). These are two different failures with two different fixes.

Nor is the cause the weights (Tab. 5). Neither recognition-pretrained encoders (PointNet++ [26], PointNeXt-S [27]) nor the deployed model's own feasibility-trained PTv3 [30] lets clearance be linearly read from its features. Only an explicit clearance tokenization of the same points does. Pretraining is irrelevant: GraspFC-PTX is near-perfect on the natural benchmark yet at 0.498 CF-pair, whether trained from scratch or from Sonata self-supervised weights [31]. Reconstruction does not help either. The deciding obstacle is a small fraction of the points, so a Point-BERT-style autoencoder [37] discards most of it on encoding. Nothing in the natural supervision ever required the score to move in the right direction, which is the sense in which the term is not extracted. Clearance supervision changes that (Sec. 4.5).

Encoder	Supervision	Δfeat.	Probe R^2
PointNet++	recognition	-0.014	0.03
PointNeXt-S	recognition	$+0.001$	0.05
PTv3	feasibility	$+0.002$	0.01
Clearance tokens (ours)	clearance	$+\mathbf{1.48}$	**0.71**

Table 5. Frozen-encoder response to a feasibility-flipping obstacle. *Supervision* is what each encoder was trained to encode (recognition on ModelNet40 [32], feasibility, or clearance); PTv3 is the deployed model's own backbone, so it is the in-domain control. Δfeat. is the pooled-feature change in native units (within-row only, *not* comparable across rows); Probe R^2 is the cross-encoder readout of how well a linear map recovers corridor clearance. Only the clearance tokenizer both responds and is readable. These are frozen encoders probed in isolation, and are not comparable to the deployed forward-pass displacements quoted in the text.

Intervention	Nat.	CF-pair	Verdict
More natural data	0.990	0.498	term not identified
Fine-tune deployed	0.990	0.495 to 0.508	does not transfer
Train from scratch on CF	0.990	**0.991**	fits intervention

Table 6. More data does not teach a transferable collision term; the from-scratch cure fits the intervention and is placement-bound (Fig. 3).

4.5. What does not close the gap, and what does

Nothing on the natural benchmark closes the gap (Tab. 6). Scaling natural data does nothing. Fine-tuning the deployed model on counterfactuals does not transfer across five standard recipes, and width is not the lever either, since a $10\times$ wider encoder tracks the baseline. The shortcut is imprinted in the pretrained weights. The obvious escape is to retrain the same architecture from scratch on counterfactual labels. That does raise the in-distribution counterfactual score (CF-pair 0.991), which alone shows the limit is not representational in principle. But it is *fitting the intervention* rather than learning clearance. The cured model is placement-bound, and its score falls back toward chance the moment the obstacle leaves the trained corridor (Fig. 3), so retraining on a new placement only fits a new shortcut. Counterfactual data buys coverage of the trained obstacle distribution, not a transferable collision representation. The lever is the representation, which we turn to next.

Clearance supervision on the unmodified benchmark. Concretely, the collision term can be learned at train time from the unmodified benchmark alone, with no counterfactual scenes, by exposing its geometric signal (corridor clearance) to the network. Fig. 7 shows the ladder, all trained

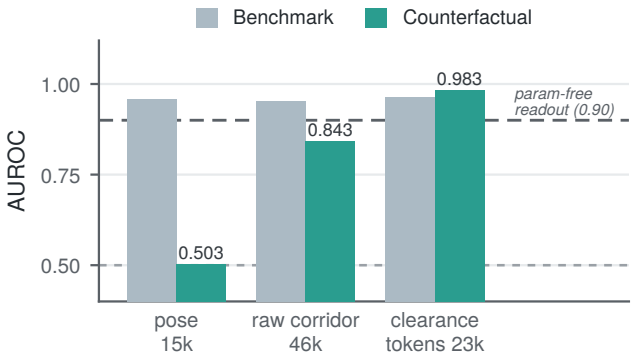


Figure 7. Train-time learning on the unmodified benchmark (plain BCE, no counterfactual scenes). Three representations of the same corridor points, all high on the benchmark; only the clearance-aware tokenization separates under the protocol (CF-pair 0.983). Dashed: the param-free geometric readout.

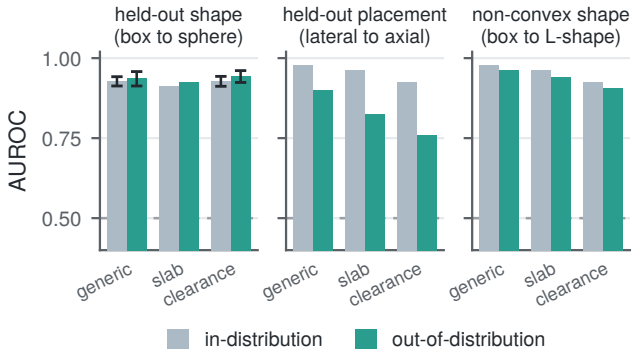


Figure 8. Out-of-distribution transfer of the three cured models of Sec. 4.5 (generic, slab, clearance). Each panel is a held-out obstacle family, planner-relabeled; bars are AUROC in- versus out-of-distribution and the dashed line is chance (0.5). Cures transfer at or near their in-distribution score; box-to-sphere error bars are bootstrap 95% CIs (n=532 ID, 250 OOD).

on the original splits with plain BCE. A pose representation stays at chance (CF-pair 0.503). Raw corridor points recover part of the term (0.843). A clearance-aware tokenization of the *same* points reaches CF-pair 0.983 with only 23k parameters, beating a param-free geometric readout of those tokens (0.90). The tokenization slices the 10 cm corridor into K=13 axial slabs (supp. Fig. 17). This model is trained on the unmodified benchmark, and is distinct from the from-scratch 0.991 of Tab. 6, which trains on *counterfactual* data. Two further train-time changes sharpen it on the same data: a per-point clearance channel, and a soft-min over per-region clearances that makes the predictor monotone by construction (supp. Appendix G). Two models tied on the benchmark thus separate under the protocol on no new data. The separation comes from a *learned* read of the clearance tokens, not an analytic check bolted onto the output. This is a diagnostic, not a deployable predictor. Like the from-scratch cure it still degrades off the trained placement distribution, as we show next, so a clearance representation that *generalizes* across placement (supp. Appendix G) is future work.

Transfer to unseen obstacle shapes and placements. The hostile reading is that a counterfactual-cured model merely memorized our obstacle generator. It did not: trained on box counterfactuals and evaluated, planner-relabeled, on never-seen *spheres*, the cure transfers at or above its in-distribution score, while a no-cloud control stays at chance. It also holds on a never-seen *non-convex* L-shape and transfers, more weakly, to an unseen on-axis *placement* family, in every case far above chance. The cure computes occupancy from the points, so it transfers across held-out obstacle shape and, more weakly, placement, rather than memorizing one generator. Per-architecture scores and bootstrap CIs are in Fig. 8.

5. Discussion and Conclusion

We audit grasp-feasibility predictors with the *corridor insertion*: a planner-verified counterfactual edit that holds the grasp pose fixed, places a solid obstacle in the approach corridor, and enters its points into the model input. It flips the label on every pair and survives the model's downsampling. Collision decides 64.3% of infeasibility, but predictors with near-perfect natural AUROC score 0.47 to 0.504 on the counterfactual pairs. The obstacle does reach the representation: in GraspFC-NNet its scene feature moves by 21%. The score still does not change with whether the obstacle blocks. The natural distribution almost never contains the fixed-pose contrast that would require it to (Sec. 4.1). The input carries the collision term, and the benchmark does not identify it.

The failure is not uniform. GRN reads a bounding box, SceneCollisionNet a 12.5 cm voxel, GraspFC-NNet registers the obstacle but cannot say whether it blocks, and GraspFC-PTX, the deployed model, does not register it at all ($|\Delta p| \approx 0.001$). Each fails below its own resolution. This holds across placement, dense clutter, the real-object control, and every planner-swap condition. Clearance supervision closes the gap: the benchmark's own labels are sufficient, and a 23k-parameter read of corridor clearance reaches CF-pair 0.983 (Sec. 4.5), while training on the interventions instead only fits them. That read stays bound to its trained placement distribution, and we evaluate in simulation with a sampling-based planner as oracle. We release the pairs, metrics, datasheet, and pipeline, so new predictors can be audited without a planner call (supp. Appendix I). Generalizing a clearance representation across obstacle placement remains an open problem.

Acknowledgments

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References

[1] Smail Ait Bouhsain, Rachid Alami, and Thierry Simeon. Extending task and motion planning with feasibility prediction: Towards multi-robot manipulation planning of realistic objects. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[2] Smail Ait Bouhsain, Rachid Alami, and Thierry Simeon. Learning geometric reasoning networks for robot task and motion planning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Sajid Ansari, Arthi, Girish Varma, and Antony Thomas. Learning motion feasibility from point clouds in cluttered environments. *arXiv preprint arXiv:2606.26700*, 2026.

[4] Erwin Coumans and Yunfei Bai. PyBullet, a Python module for physics simulation for games, robotics and machine learning. http://pybullet.org, 2016–2021.

[5] Michael Danielczuk, Arsalan Mousavian, Clemens Eppner, and Dieter Fox. Object rearrangement using learned implicit collision functions. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[6] Neil T. Dantam, Zachary K. Kingston, Swarat Chaudhuri, and Lydia E. Kavraki. An incremental constraint-based framework for task and motion planning. *International Journal of Robotics Research, Special Issue on the 2016 Robotics: Science and Systems Conference*, 37(10):1134–1151, 2018.

[7] Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[8] Congyue Deng, Or Litany, Yueqi Duan, Adrien Poulenard, Andrea Tagliasacchi, and Leonidas J. Guibas. Vector neurons: A general framework for SO(3)-equivariant networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[9] David Emukpere, Romain Deffayet, and Jean-Michel Renders. Robust skills, brittle grounding: Diagnosing restricted generalization in vision-language action policies via multi-object picking. *arXiv preprint arXiv:2602.24143*, 2026.

[10] Clemens Eppner, Arsalan Mousavian, and Dieter Fox. ACRONYM: A large-scale grasp dataset based on simulation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[11] Hao-Shu Fang, Chenxi Wang, Minghao Gou, and Cewu Lu. GraspNet-1Billion: A large-scale benchmark for general object grasping. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[12] Matt Gardner, Yoav Artzi, Victoria Basmova, et al. Evaluating models' local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP*, 2020.

[13] Caelan Reed Garrett, Tomás Lozano-Pérez, and Leslie Pack Kaelbling. FFRob: Leveraging symbolic planning for efficient task and motion planning. *The International Journal of Robotics Research*, 37(1):104–136, 2018.

[14] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12), 2021.

[15] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019.

[16] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2, 2020.

[17] Leslie Pack Kaelbling and Tomás Lozano-Pérez. Integrated task and motion planning in belief space. *The International Journal of Robotics Research*, 32(9-10):1194–1227, 2013.

[18] Sertac Karaman and Emilio Frazzoli. Sampling-based algorithms for optimal motion planning. *The International Journal of Robotics Research*, 30(7):846–894, 2011.

[19] Divyansh Kaushik, Eduard Hovy, and Zachary C. Lipton. Learning the difference that makes a difference with counterfactually-augmented data. In *International Conference on Learning Representations (ICLR)*, 2020.

[20] J. Chase Kew, Brian Ichter, Maryam Bandari, Tsang-Wei Edward Lee, and Aleksandra Faust. Neural collision clearance estimator for batched motion planning. In *International Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2020.

[21] James J. Kuffner and Steven M. LaValle. RRT-Connect: An efficient approach to single-query path planning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2000.

[22] R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.

[23] Adithyavairavan Murali, Arsalan Mousavian, Clemens Eppner, Adam Fishman, and Dieter Fox. CabiNet: Scaling neural collision detection for object rearrangement with procedural scene generation. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.

[24] Rhys Newbury, Morris Gu, Lachlan Chumbley, Arsalan Mousavian, Clemens Eppner, Jürgen Leitner, Jeannette Bohg, Antonio Morales, Tamim Asfour, Danica Kragic, Dieter Fox, and Akansel Cosgun. Deep learning approaches to grasp synthesis: A review. *IEEE Transactions on Robotics (T-RO)*, 2023.

[25] Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[26] Charles R. Qi, Li Yi, Hao Su, and Leonidas J. Guibas. PointNet++: Deep hierarchical feature learning on point sets in a

metric space. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[27] Guocheng Qian, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Abed Al Kader Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. PointNeXt: Revisiting PointNet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[28] Martin Sundermeyer, Arsalan Mousavian, Rudolph Triebel, and Dieter Fox. Contact-GraspNet: Efficient 6-DoF grasp generation in cluttered scenes. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[29] Antony Thomas, Fulvio Mastrogiovanni, and Marco Baglietto. MPTP: Motion-planning-aware task planning for navigation in belief space. *Robotics and Autonomous Systems*, 141:103786, 2021.

[30] Xiaoyang Wu, Li Jiang, Peng-Shuai Wang, Zhijian Liu, Xihui Liu, Yu Qiao, Wanli Ouyang, Tong He, and Hengshuang Zhao. Point transformer V3: Simpler, faster, stronger. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[31] Xiaoyang Wu, Daniel DeTone, Duncan Frost, Tianwei Shen, Chris Xie, Nan Yang, Jakob Engel, Richard Newcombe, Hengshuang Zhao, and Julian Straub. Sonata: Self-supervised learning of reliable point representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[32] Zhirong Wu, Shuran Song, Aditya Khosla, Fisher Yu, Linguang Zhang, Xiaoou Tang, and Jianxiong Xiao. 3D ShapeNets: A deep representation for volumetric shapes. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.

[33] Youguang Xing, Xu Luo, Junlin Xie, Lianli Gao, Hengtao Shen, and Jingkuan Song. Shortcut learning in generalist robot policies: The role of dataset diversity and fragmentation. In *Conference on Robot Learning (CoRL)*, 2025.

[34] Lei Xu, Tianyu Ren, Georgia Chalvatzaki, and Jan Peters. Accelerating integrated task and motion planning with neural feasibility checking. *arXiv preprint arXiv:2203.10568*, 2022.

[35] Zhutian Yang, Caelan R. Garrett, Tomás Lozano-Pérez, Leslie Kaelbling, and Dieter Fox. Sequence-based plan feasibility prediction for efficient task and motion planning. In *Robotics: Science and Systems (RSS)*, 2023.

[36] Chenning Yu and Sicun Gao. Reducing collision checking for sampling-based motion planning using graph neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[37] Xumin Yu, Lulu Tang, Yongming Rao, Tiejun Huang, Jie Zhou, and Jiwen Lu. Point-BERT: Pre-training 3D point cloud transformers with masked point modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Auditing Grasp Feasibility Predictors with Planner-Verified Counterfactuals

Supplementary Material

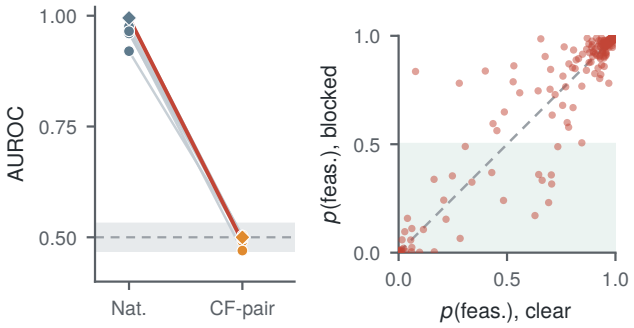


Figure 9. *Left*: every predictor in Tab. 1 falls from near-perfect natural AUROC to near 0.5 under the same-pose counterfactual (one line each; GraspFC-NNet red). *Right*: for GraspFC-NNet, the blocked grasp gets essentially the clear grasp's score, so the same-pose pairs whose label flips lie on the $y{=}x$ diagonal rather than in the region a collision-aware model would occupy (shaded).

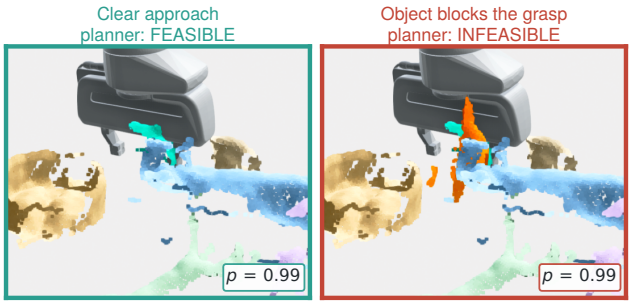


Figure 10. The obstacle is a GraspNet-1B object (from the scene segmentation, same Novel split) resting on the table next to the target. The planner relabels the grasp infeasible; the deployed SOTA returns the identical score. Over 150 such pairs the planner flips every one while the model cannot tell the two apart (CF-pair AUROC 0.5): the result extends to dataset objects, not only inserted boxes.

Roadmap. We first give the robustness figures (oracle standoff, obstacle type, a dataset object, and dense clutter) and the physical cost of the result; Sec. A then extends the protocol to multi-step plan feasibility; Sec. B gives the shape- and contact-resolution detail for GRN and SceneCollisionNet on their own benchmarks; Sec. C lists per-model architecture, training pool, and in-distribution accuracy; Sec. D gives the per-model interventional scores and the representation-to-failure taxonomy; Sec. E dissects why GRN partially passes; Sec. F reports the negative controls and label-integrity checks; Sec. G specifies the clearance-aware tokenization; and Secs. H and I provide the datasheet and the audit protocol for the released benchmark.

Robustness of the diagnosis: figures and the physical cost

The main paper establishes that the diagnosis survives the oracle standoff, the obstacle type, the planner, and the arm (Tab. 3). Here we give the figures behind that result, the per-condition planner-swap scores, a worked example, dense clutter, and the downstream cost of the result.

A worked example. Fig. 2 in the main paper shows one such instance in full: a single grasp on a GraspNet-1B scene under all four conditions. Tab. 8 reports its per-condition scores. The three obstacles trigger two distinct geometric failure modes (a blocked approach for the corridor box and the resting pillar, a blocked transit for the path box), and the deployed SOTA returns the same $p(\text{feasible})$ to within 0.001 across all four, although the planner flips three of them.

Robot	Planner	Agree	κ	Flip
Panda	BiRRT-Connect, re-run (control)	0.995	0.990	**1.000**
Panda	BiRRT-Connect, other hyper-params	0.990	0.977	**1.000**
Panda	single-tree goal-biased RRT	0.973	0.940	**1.000**
Panda	Lazy-PRM	0.992	0.983	**1.000**
xArm6	BiRRT-Connect, re-run (control)	0.988	0.973	**1.000**
xArm6	BiRRT-Connect, other hyper-params	0.989	0.975	**1.000**
xArm6	single-tree goal-biased RRT	0.984	0.964	**1.000**
xArm6	Lazy-PRM	0.989	0.975	**1.000**

Table 7. What the causal edit does to the label, under four planning algorithms and two kinematic chains. The labels themselves are not identical across conditions; the flip is. Only the planner object or the robot changes; the scene, the grasp poses, the contact tolerance, and the 21-step approach discretization are the same code. *Agree* is label agreement with the released oracle and κ is Cohen's κ, both over the clear, causal, and null conditions. *Flip* is the fraction of causal edits the planner relabels infeasible. The causal edit flips the label in 8 of 8 conditions without exception, over 8,412 planner calls. Residual disagreement sits on the clear and null branches and is bounded by the RNG-only control in the first row of each block. The xArm6 is the harder embodiment: 607 of 957 Panda-feasible candidate grasps are out of its reach, and every corridor insertion still flips the label.

This is not a hand-selected pair: it is one of the instances counted in Tab. 3, where the planner flipped every obstacle type while the model never moved.

Condition	Planner	Failure mode	model p
Clear approach	feasible	none	0.975
Floating corridor box	infeasible	blocked approach	0.976
Resting pillar	infeasible	blocked approach	0.975
Transit-path box	infeasible	blocked transit	0.975

Table 8. Per-condition scores for the worked example of Fig. 2: one grasp under four conditions (deployed SOTA, released weights).

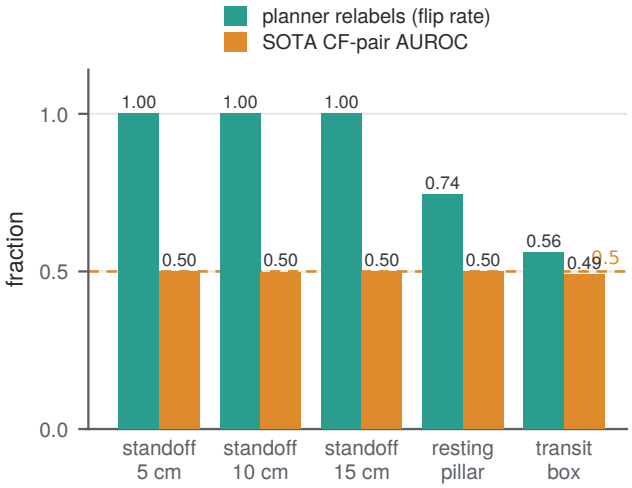


Figure 11. Tab. 3 as a chart: planner relabel rate (teal) versus deployed-SOTA CF-pair AUROC (amber) across three standoffs and two further obstacle types (a gravity-resting pillar and a transit-path box). The dashed line at 0.5 applies only to the amber AUROC, not to the teal flip rate.

Dense multi-object clutter. The diagnosis also survives dense clutter rather than a single inserted body. Inserting a cluster of four solid obstacles into the approach flips all 150 test grasps the planner re-checks (causal flip 1.00, across two failure modes: 149 blocked-approach and 1 blocked-transit), yet the deployed SOTA returns the same $p(\text{feasible})$ with and without the cluster (CF-pair AUROC 0.498, $|\Delta p| < 0.001$); Fig. 12 shows one example.

The result has a physical cost. The score gap is not only a ranking artifact; it changes what a robot does. On one counterfactual scene (Novel split, scene 160) we execute the candidate grasps in physics simulation (Fig. 13). The pose-shortcut model's top-ranked grasp, the one it scores feasible, drives the gripper into the inserted corridor box, a pybullet-verified contact. Screening the same 20 candidate grasps with the clearance-aware read of Sec. 4.5 instead returns a top-1 whose approach is collision-free, verified by a BiRRT path. This is one scene, illustrating the score's downstream effect, not a deployment claim. Same scene, same candidates: the deciding difference is whether the score reads the geometry the planner checks.

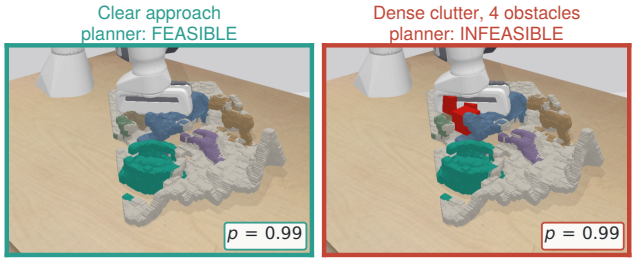


Figure 12. Dense multi-object clutter on a GraspNet scene: a cluster of four solid obstacles inserted into the approach makes the planner relabel the grasp infeasible (right), while the deployed SOTA returns the same $p(\text{feasible})$ as the clear scene.

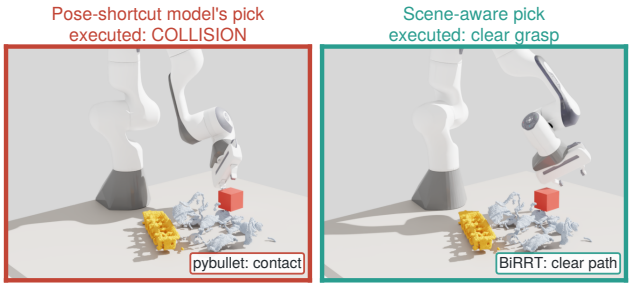


Figure 13. Closed-loop safety demo on one counterfactual scene (Novel split, scene 160), path-traced in Isaac Sim by replaying the pybullet link poses. *Left*: the pose-shortcut model's top pick drives the gripper into the inserted box (pybullet-verified collision). *Right*: the clearance-aware screen of the same 20 candidates returns a collision-free top-1 (BiRRT-verified).

A. The protocol extends to multi-step plan feasibility

Single-grasp feasibility is one action; many planners predict feasibility for a whole action *sequence*. Our protocol applies unchanged (Fig. 15): insert an obstacle, re-run the oracle on the edited state, and relabel each step and the suffix it gates. Two points make this concrete. **(i)** PIGINet [7], one of the audited models, is itself a sequence-based plan-feasibility predictor; run from its released weights on its authors' own PR2 kitchen it calls a planner-blocked plan feasible 90% of the time (Fig. 14; Fig. 4, Sec. 4.3), so the shortcut is already demonstrated on a multi-step predictor, not only on single-grasp classifiers. **(ii)** Our release includes a whole-skeleton feasibility generator that carries symbolic preconditions and accumulated geometric state across k steps, so the same insert-and-relabel protocol extends to sequences with no change to the metric.

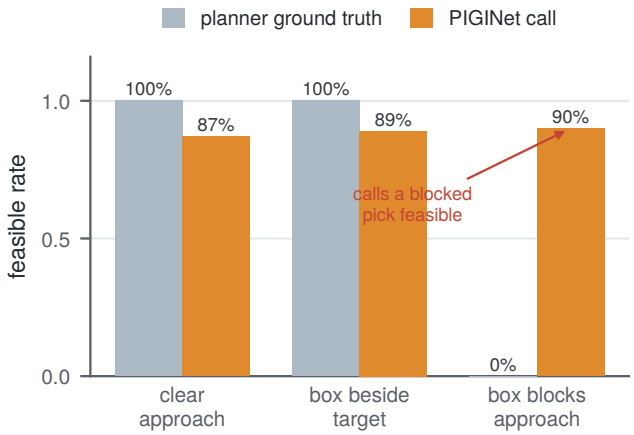


Figure 14. PIGINet on its own PR2 kitchen (released weights), aggregated over 450 counterfactual pairs. Its binarized feasibility call is flat (87 to 90%) across a clear approach, a harmless box beside the target, and a box blocking the approach, while the planner ground truth drops to zero for the blocking case.

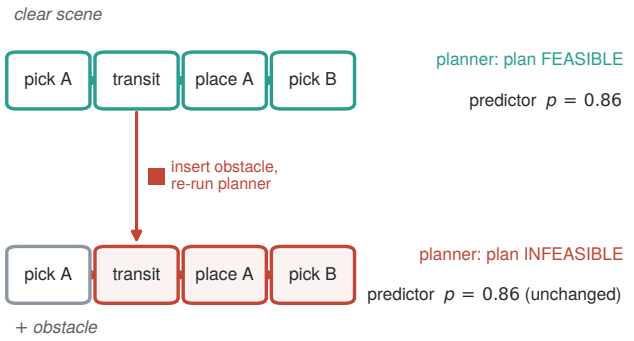


Figure 15. The protocol at the plan level. A multi-step rearrangement plan is plan-feasible when the scene is clear (top); inserting an obstacle at one step makes the planner relabel that step and every step after it infeasible (bottom), yet a plan-feasibility predictor returns the same score.

B. Shape and contact resolution on the methods' own data

Fig. 6 shows the recreated GRN scene in the main paper; here we give the quantitative detail of the two released-weights results that are easiest to misread, and the recreated SceneCollisionNet scenes (Fig. 5). **GRN** detects an inserted obstacle on its panda_3d_4 benchmark (feasibility 0.99 clear, 0.94 beside, 0.01 on-path), but a box and its inscribed same-bounding-box ellipsoid receive an identical score, so the shape-swap AUROC is exactly 0.500 and shape is decisive for 14% of configurations (Fig. 16): GRN reads the bounding box, not the shape. **SceneCollisionNet**, run from the released checkpoint on an ACRONYM-style cluttered tabletop in its trained workspace, does not

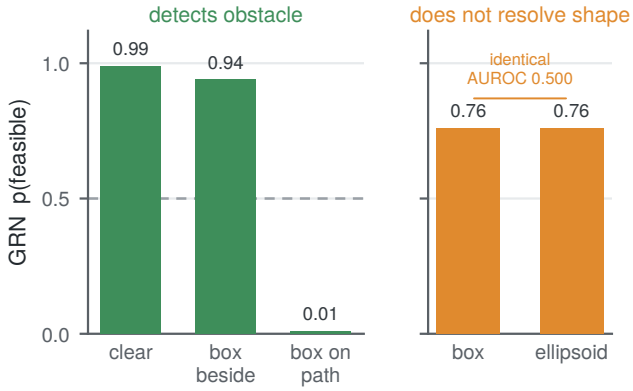


Figure 16. GRN on its panda_3d_4 benchmark, released weights. Left: GRN detects an inserted obstacle (feasibility drops only when the box is on the path). Right: a box and its inscribed same-bounding-box ellipsoid get an identical score (shape-swap AUROC 0.500).

resolve its own contact boundary: as a gripper-held cube is swept against a solid obstacle from overlapping it to a 6 cm clear gap (Fig. 5), the released score stays pinned at $p_{\text{coll}} = 0.76$ (full-sweep range 0.75 to 0.77, never below 0.5), scoring a deep collision and a 6 cm clearance identically where an ideal checker steps from 1 to 0. At its 12.5 cm voxel it is geometry-agnostic below the cell size: in controlled box probes it registers gross presence (removing the obstacle drops the score 0.67 to 0.12) but resolves shape only weakly (same-bounding-box shape swap, 0.821 AUROC, not a decision boundary). The common factor across all three released-weights systems is geometry-agnosticism below each representation's resolution.

C. Per-model training and architecture

Every re-implemented predictor is trained on the same GraspNet-1B planner labels under one protocol (shared oracle, splits, and a binary-cross-entropy objective); the three GraspFC classifiers are the feasibility predictors of [3], and GRN is run from released weights on its own data. Tab. 9 gives each model's representation, training pool, and in-distribution accuracy, so a reader can see that the interventional gap of Tab. 1 is not an under-training artifact: every model reaches high in-distribution AUROC on the same labels before its score falls under the intervention. Class balance is the benchmark's natural 24.1% feasible on the 100-scene training split (the full label pool is 9.6% feasible); PIGINet's kitchen pool is 69% feasible, close to the authors' own 67%. The three GraspFC classifiers share one recipe (binary cross-entropy with logits, positive weight 3.14, AdamW at 1.5×10^{-3} with cosine decay, 30 epochs, batch 128) on the benchmark's 100-scene, 1.52M-grasp training split; each re-implementation follows its own pa-

per's loss, noted in Tab. 9.

D. Per-model interventional results and the representation-to-failure taxonomy

Tab. 1 reports natural and CF-pair AUROC; here we read it two ways. First, the representation-to-failure taxonomy (Tab. 10) names, for each model, the coarsest collision geometry its representation resolves and the mechanism behind the gap; the single common axis is that each model is geometry-agnostic below its representation's resolution. Second, the behavioral gallery (Tab. 11) gives the score each model returns *with and without* the flip-causing obstacle on a matched set: five predictors return essentially the clear-scene score ($|\Delta p| \leq 0.02$) while the planner flips the label, and GRN, the one model with a supervised obstruction head fed the box, moves with the planner. The obstacle is verified present in every model's input (it renders into NFC's dedicated neighbor channel, into all five of AGFP-Net's depth views, and, for the deployed SOTA, a frame audit re-places it on the approach corridor in the model's own input frame; Sec. F), so a flat score is the representation discarding visible evidence, not a missing input.

E. Why GRN partially passes, and where it does not

GRN is the one audited model that reacts to an inserted obstacle, so it deserves a careful reading: it is neither a counterexample to the diagnosis nor a solved case. The difference is structural. GRN has an explicit, supervised, factored grasp-obstruction head that is fed each object's box and pose, which the monolithic predictors lack, and on the two things that head can express it is causally faithful. **(i) Presence.** Removing the true obstructor flips GRN to feasible $94/91/90\%$ of the time across $4/10/20$-object scenes, while removing a random non-obstructor flips it 0%, ruling out an any-edge-removal artifact. On the same insertion probe we apply to the other models, GRN scores clear versus blocked AUROC 1.000 ($p\,0.990 \rightarrow 0.007$; a beside control stays at 0.943): the same probe that leaves the other models at chance moves GRN end to end. Instrumenting GRN's graph construction confirms the inserted node enters the computation and its obstruction head fires proportionally to geometry (1.00 in the approach versus 0.28 beside), so the discriminator is the head's reasoning, not mere edge presence. **(ii) Bounding-box clearance.** Shrinking or sliding the obstacle moves GRN's score smoothly and monotonically, crossing 0.5 right as the boxes separate; it tracks clearance magnitude, not just presence, at the bounding-box level.

Where it fails is below that representation. GRN's per-object input is an axis-aligned box plus yaw, with no shape and no out-of-plane pose, so any geometry the box cannot encode is invisible by construction, and the failures span both error directions (Tab. 12). A solid box and its inscribed same-bounding-box ellipsoid receive an identical prediction, yet 14% of configurations are shape-decisive (the box intersects the approach corridor while the ellipsoid does not), giving a shape-swap AUROC of exactly 0.500. A center-fixed obstacle tilted into the approach is identical to the same object standing upright, because pitch and roll are dropped at the node, so on 72.5% of these tilted-obstacle cases GRN reads the object as upright and commands a colliding grasp: a false-clear, the unsafe direction. An open frame or bin is filled to its solid bounding box, so GRN abandons a reachable grasp 100% of the time: a false-block. Each is a matched pair GRN scores identically while true feasibility flips, so its AUROC is chance on all three against an oracle's 1.0.

For a datasets venue, these geometries are out of GRN's box-only benchmark distribution, and that is the point: GRN's high score on its own benchmark is partly due to the benchmark's limited geometry (axis-aligned boxes), and the counterfactuals expose the failures the dataset hides. This is a dataset-and-representation critique, not a claim that GRN is broken. (A practitioner could give GRN the recomputed bounding box of a tilted object, but that is external geometry GRN does not compute, and it would then over-block; the native representation assumes upright boxes.)

F. Negative controls and label integrity

The diagnosis rests on the flip being caused by the true obstacle and on that obstacle being present in the model's input, so we report the controls that close the obvious alternative explanations.

The obstacle is on the corridor in the model's own frame. A concern is that the obstacle might float off the approach in the model's input frame, which would make $\Delta p \approx 0$ a placement artifact. A frame audit re-places the box on the corridor in the deployed SOTA's input frame and re-scores: the model still does not respond (clear-versus-box AUROC $0.498 \rightarrow 0.501$, $\Delta p \approx 0$), so the result is not a misplacement artifact.

Flips are caused by the true obstructor, not any edit. A matched box placed 30 cm off-corridor leaves the grasp feasible 98.0% of the time, and the same holds for a box slid laterally off the finger axis (97% feasible); for GRN, removing a random non-obstructor flips the label 0% of the time versus 94% for the true obstructor. The edits are causal and specific.

Non-diagnostic perturbations are excluded by design. A perturbation is only informative if the oracle relabels it. Naive cloud edits that do not change the label (50% point dropout, 5 to 10 mm jitter) move every model's AUROC by 0.000, which shows only that the model is insensitive to the cloud, not that it is wrong; we therefore never count them,

Model	Representation (what enters the net)	Training pool	Par.	Objective	Authors' reported (own data)	Nat.
GraspFC-NNet	pose + 0.3 m grasp-local crop, pooled MLP	benchmark train split	1.65M	BCE	[3]	0.995
GraspFC-Conv3D	pose + voxel grid, 3D CNN	benchmark train split	9.7M	BCE	[3]	0.976
GraspFC-PTX	pose + point cloud, PTv3 backbone	benchmark train split	9.0M	BCE	[3]	0.996
SceneCollisionNet	per-point MLP, scatter-max to 12.5 cm voxel, 3D conv; PointNet++ gripper	18k labels	26M	hard-neg BCE	acc 93.2%, AP 0.990 [4]	0.964
PIGINet	per-object CLIP ViT-B/32 image + pose token, 4 attention blocks	12k labels	31M	pos-weighted BCE	val acc 0.89, 80% plan-time cut [7]	0.960
NFC	object-centered 2-ch depth 100×100 (ch. 2 = neighbors), 2 conv + 4 FC	30k labels	22.8k	weighted BCE	acc 93.6%, 63.5% time saved [6]	0.965
AGFP-Net	5 orthographic depth views + target mask, 3× ResNet-18	18k labels	33M	BCE	plan success 90 to 100% [1]	0.920
GRN (released)	box $[l, w, h, x, y, z, \text{yaw}]$ graph, GNN + supervised obstruction head	panda_3d_4 (20k scenes), released weights	2.3M	(released)	F1 0.94/0.94/0.99, MAE 0.028 [2]	0.995

Table 9. Per-model representation, training pool, and in-distribution accuracy. *Nat.* is the shared-benchmark natural AUROC of Tab. 1 (GRN on its own panda_3d_4 task); *Authors' reported* is each method's published headline on its own data, none of which is AUROC, which is why we re-measure all of them under one Nat. The re-implementations are trained on planner labels under one protocol; PIGINet loads its released head and the GRN run matches the paper's F1/MAE.

Model	Resolves at	Nat.	CF-pair	Failure mechanism
GraspFC-NNet	crop pool	0.995	0.491	registers the obstacle; feature displacement is direction-random
GraspFC-Conv3D	voxel grid	0.976	0.497	obstacle in the voxel grid, discarded
GraspFC-PTX	point pool	0.996	0.498	obstacle in the cloud, discarded
SceneCollisionNet	12.5 cm voxel	0.964	0.470	contact boundary below the 12.5 cm voxel
PIGINet	object token	0.960	0.500	one object token, no clearance encoded
NFC	2-ch depth	0.965	0.504	predicts from pose, does not use neighbor channel
AGFP-Net	5-view depth	0.920	0.498	obstacle in all 5 views, discarded
GRN (released)	box + yaw	0.995	0.500*	detects box; does not resolve sub-box shape (Sec. E)

Table 10. Representation-to-failure taxonomy. "Resolves at" is the coarsest collision geometry each representation can express; Nat. and CF-pair are from Tab. 1. *GRN moves on presence (it has a supervised obstruction head fed the box), so its entry is the same-bounding-box shape swap, dissected in Sec. E.

Model	clear p	obstacle p	Δp
GraspFC-PTX	0.947	0.947	0.000
SceneCollisionNet	0.70	0.72	+0.02
PIGINet	0.766	0.766	0.000
NFC	0.798	0.797	−0.001
AGFP-Net	0.702	0.701	+0.001
GRN (released)	0.990	0.007	−0.983

Table 11. Behavioral gallery: predicted $p(\text{feasible})$ on a matched set before and after inserting the flip-causing obstacle, which the planner relabels feasible→infeasible in every row. Values are means over the matched set; the worked example of Fig. 2 (0.975) is one such pair.

Hole	GRN AUROC	Mislabels
Shape	0.500	14% (false block)
Tilt	0.500	72.5% (false clear)
Concavity	0.500	100% (false block)

Table 12. GRN passes presence and bounding-box clearance but is at chance (AUROC 0.5) for geometry finer than its bounding box: three sub-bounding-box holes (shape, tilt, concavity), each a matched pair GRN scores identically while the corridor oracle flips. "Mislabels" is the fraction of probed targets the hole gets wrong; the failures span both error directions (false block and false clear).

and use only the insert-and-relabel intervention.

Labels are effectively deterministic. Re-running the planner with 8 seeds on held-out grasps gives $0/480$ flips (label noise $\leq 0.6\%$ at 95% confidence), so the 0.99-level natural scores are not noise-limited.

The natural label is pose-recoverable, which is why natural data cannot supervise the term. A 25-nearest-neighbor lookup on grasp pose alone, with no scene at all, scores 0.963 natural AUROC on SceneCollisionNet's task. The natural distribution carries the feasibility label in the pose, so a model is never forced to read the scene, which is the standing reason the counterfactual contrast is necessary rather than convenient.

The in-crop gap is representational, not an information limit. When the obstacle leaves the input crop (a distal control 30 cm back), the cures fall to chance (raw 0.517, clearance tokens 0.505): an information limit, distinct from the in-crop case where the obstacle is present but discarded. The global-input cure (from-scratch on counterfactuals, full cloud) partially survives the distal move (0.602, versus 0.497 for the as-is deployed model, Tab. 2), confirming the in-crop drop is about the representation, not the input.

G. The clearance-aware tokenization (architecture and scope)

Sec. 4.5 shows that the collision term, which no audited predictor resolves at the scale that decides the collision, is recoverable at train time from the original benchmark by a clearance-aware read of the grasp-local points. This appendix gives the construction behind that result (Fig. 17) and states what it does and does not establish. It is a *diagnostic*: it isolates *where* the missing signal lives (the representation), not a deployable collision predictor. The distinction matters because, as Sec. 4.5 shows, a learned read of these tokens is still placement-bound, like the from-scratch counterfactual cure.

Tokenization. We work in the gripper frame, where collision is exactly SE(3)-invariant (the obstacle's gripper-frame coordinates are unchanged by any rigid motion of the scene-plus-grasp). We slice the 10 cm approach corridor into $K = 13$ axial slabs along the approach axis a. Each non-target scene point is projected onto a to get an axial coordinate z and a lateral radius ρ (its perpendicular distance from the approach line); points outside the corridor extent are dropped. Slab k emits a three-number token $[\,\rho_k,\ \mathrm{occ}_k,\ z_k\,]$: ρ_k is the minimum lateral clearance of any obstacle point in that slab (capped), occ_k is its occupancy (a soft point count), and z_k is the slab center. The result is a fixed $K \times 3$ tensor regardless of how many points fall in the corridor, so the representation is point-count-agnostic and obstacle-localizing by construction: the channel ρ_k is a direct readout of where the swept gripper comes closest to matter, the very quantity the planner checks.

Network. A shared per-slab MLP ($3 \to 32$, GELU) maps each token to an embedding; we max-pool over slabs and append the two global statistics $\min_k \rho_k$ and $\max_k \mathrm{occ}_k$, then a small collision MLP produces a clearance feature h_C. A separate pose head produces h_R from the same 17-dimensional pose descriptor the baselines use, and the two are concatenated into the final feasibility head. This is the $R \otimes C$ factorization of Fig. 17: h_R carries reachability (pose, not SE(3)-invariant) and h_C carries collision (gripper-frame, invariant). The whole network is 23k parameters; the per-slab MLP is shared across slabs, so the parameter count is independent of K (a sweep over $K \in \{5, 9, 13, 21, 31\}$ leaves it at 22.8k and within-grasp AUROC at 0.996 to 1.0).

Two readouts, one representation. The point of the construction is that the collision signal is present in the tokens *before* any learning. Thresholding the single global statistic $\min_k \rho_k$, with no trained weights and no labels, already gives CF-pair AUROC 0.899 and within-grasp 0.911. The learned head on the identical tokens reaches CF-pair 0.983. Learning therefore *sharpens* a signal the representation already exposes; it does not manufacture it. This is the central contrast: a pose representation cannot be made to read collision at any accuracy (Tab. 5, Fig. 7), whereas a clearance representation makes the term readable at 0.90 with zero fitting. The gap is the representation, not learnability.

Two train-time innovations on the original data. Two structural changes sharpen this clearance read on the original benchmark, with no counterfactual scenes, and are the constructions behind the "sharpen" claim of Sec. 4.5. **(i) A per-point clearance channel.** Appending each point's corridor clearance to its input features raises a raw encoder's clearance-probe R^2 from 0.05 to 0.22; distilling the same signal into the *pooled* feature instead leaves it at 0.02, so the clearance must enter before pooling. **(ii) A soft-min over per-region clearances.** Replacing max-pooling with a soft-min over the per-slab clearances makes the feasibility head monotone in clearance by construction and cuts the insertion-monotonicity violation rate, the fraction of random point insertions that *raise* predicted feasibility, from 0.42 under max-pooling to 0.025. Both act at train time on the original splits, so the collision term is sharpened without any counterfactual data.

What it does not establish. The learned read is *not* a transferable collision model. Trained on one corridor placement it transfers across held-out obstacle *shape* and *size* in

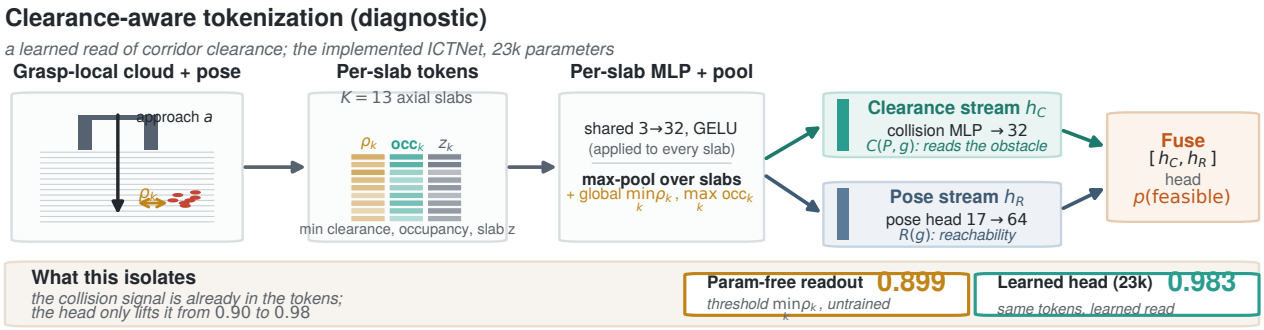


Figure 17. The clearance-aware tokenization of Sec. 4.5, the implemented 23k-parameter network. The grasp-local cloud and pose enter in the gripper frame; the 10 cm approach corridor is sliced into $K = 13$ axial slabs, each emitting a three-number token (minimum lateral clearance ρ_k, occupancy, slab center). A shared per-slab MLP, a max-pool over slabs, and the global $\min_k \rho_k$ form a clearance feature h_C fused with a pose feature h_R (the $R \otimes C$ factorization).

ranking (sphere, non-convex L, and $1.5/3/5$ cm boxes all transfer at or above the in-distribution score), but its absolute calibration drifts under size shift and its accuracy falls on an unseen on-axis *placement* family (0.74, Sec. 4.5), because workspace clearance equals configuration-space clearance only for a point robot. A transferable solution must learn a calibrated clearance *field* with the structural guarantees a binary classifier lacks (insertion-monotonicity, size transfer by Minkowski subtraction, and a certified screening bound); we treat that as the method that follows this dataset paper, not as a claim of the present work. Likewise, holding the Franka gripper fixed, varying only the finger aperture or the on-axis approach depth does not move the planner's label (the arm body dominates the swept volume), so a true cross-gripper test needs a second arm and is out of scope here.

Out-of-distribution transfer of the cures. Fig. 8 gives the per-variant breakdown behind Sec. 4.5: three counterfactual-cured models (a generic architecture, a slab-tokenized model, and the clearance-token model), each trained on box counterfactuals and re-evaluated, planner-relabeled, on a held-out shape (sphere), placement (axial), and non-convex (L-shape) family. Every cure transfers at or near its in-distribution score; on the unseen on-axis placement the generic architecture transfers best (0.90) and the clearance one least (0.74), both far above chance, while the no-cloud control stays at 0.500 throughout.

H. Datasheet

This datasheet follows Gebru et al. [5].

Composition. The counterfactual benchmark pairs each base-feasible grasp with a planner-verified clear and obstacle-inserted variant at an identical pose, on public GraspNet-1B scenes. It includes the 1,065 same-pose CF pairs used for CF-pair AUROC, graded obstacle-offset groups (one grasp with six offsets from blocking to harmless), and the held-out generalization and robustness sets used here: held-out shape (sphere), non-convex (L-shape), placement (axial), distal and adjacent placements, obstacle size ($1.5/3/5$ cm), standoff ($5/10/15$ cm), and the resting-pillar and transit-path obstacle types, plus the 150 dataset-object pairs of Fig. 10 and the dense-clutter set of Fig. 12. Each instance stores the scene and grasp identifiers, the obstacle geometry and pose, the planner label, and the binding failure mode.

Collection and labels. Labels come from an RRT-Connect motion planner for a 7-DOF Franka Panda in PyBullet (IK at the grasp and a pre-grasp standoff, a collision-free linear approach, and a collision-free path from home). Each label is a fresh planner call on the edited scene (mean 2.51 s; median 2.37; $n = 40$). The label is effectively deterministic: re-running with 8 seeds on held-out grasps gives $0/480$ flips (label noise $\leq 0.6\%$ at 95% confidence, rule of three).

Splits and intended use. Evaluation uses the official Seen/Similar/Novel object splits, with the counterfactual tests reported on Novel objects. Intended use is auditing obstacle-awareness of feasibility predictors. The protocol is also applied to three prior-work settings (SceneCollisionNet's ACRONYM, PIGINet's PR2 kitchen, GRN's panda_3d_4) to show it is not tied to our data.

Responsible use, biases, and limitations. All labels are produced in simulation by a planner oracle; the benchmark is a diagnostic of obstacle-awareness and is *not* a deployment safety certificate, and a high score does not certify a model for use on a physical robot. The data inherit GraspNet-1B's distribution (tabletop RGB-D scenes and its object set) and our single-arm Franka Panda setup, so coverage of arms, grippers, and environments is limited. By construction the natural feasibility label is largely recoverable from the grasp pose alone (Sec. F); this is the bias the counterfactual split is designed to expose rather than a de-

fect to be corrected. Obstacles are geometrically inserted solid bodies validated by the planner relabel, not by physical settling.

Distribution and licensing. We release the planner-verified annotations, the obstacle geometry, the planner configuration, and the relabeling pipeline so others can score new models without re-running a planner and can regenerate labels for new obstacle families. The underlying RGB-D scenes remain under the GraspNet-1B license for research use, so we redistribute only the new annotations and code, not the source images. We release the new planner-verified annotations and this datasheet under CC BY 4.0 and the relabeling pipeline code under the MIT license. The annotations, datasheet, and relabeling pipeline are available at https://github.com/codewithsajid/graspfc-cf.

Maintenance, versioning, and contact. We will host the annotations, the pipeline, and a public leaderboard of interventional scores (Sec. I), version the release so scores remain comparable across updates, and document any errata or added obstacle families in a changelog.

I. Using the released benchmark

The reusable contribution is that a model's obstacle-awareness can be measured without a motion planner. To audit a new predictor M: (1) load the released same-pose pairs; (2) for each pair, have M score the clear cloud and the obstacle-inserted cloud at the *identical* grasp pose, adding the released obstacle points to M's input; (3) report CF-pair AUROC over the pairs, within-grasp AUROC over the graded-offset groups, and pick-clear accuracy. A model that reads only the grasp pose scores chance on all three by construction: on the released pairs the deployed SOTA and a pose-only baseline both score pick-clear 0.500, exactly as on CF-pair, while a clearance-aware read of the same points reaches 0.997 (raw corridor points recover part of it, 0.817), so the metrics cannot be satisfied by a pose lookup. No planner call is needed at audit time: scoring the $1{,}065$-pair head-to-head set takes under a minute on one GPU, versus the ≈ 45 minutes of planner time the labels themselves cost. Each obstacle family (shape, placement, size, stand-off, resting, transit) is a separate split on which all three metrics are reported, so a submission can show which interventions a model survives. Regenerating labels for a *new* obstacle family is the only step that calls the planner, and the released pipeline does it from a single geometry specification. We will host the annotations, pipeline, and a public leaderboard of interventional scores.

References

[1] Smail Ait Bouhsain, Rachid Alami, and Thierry Simeon. Extending task and motion planning with feasibility prediction: Towards multi-robot manipulation planning of realistic objects. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

[2] Smail Ait Bouhsain, Rachid Alami, and Thierry Simeon. Learning geometric reasoning networks for robot task and motion planning. In *International Conference on Learning Representations (ICLR)*, 2025.

[3] Sajid Ansari, Arthi, Girish Varma, and Antony Thomas. Learning motion feasibility from point clouds in cluttered environments. *arXiv preprint arXiv:2606.26700*, 2026.

[4] Michael Danielczuk, Arsalan Mousavian, Clemens Eppner, and Dieter Fox. Object rearrangement using learned implicit collision functions. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.

[5] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12), 2021.

[6] Lei Xu, Tianyu Ren, Georgia Chalvatzaki, and Jan Peters. Accelerating integrated task and motion planning with neural feasibility checking. *arXiv preprint arXiv:2203.10568*, 2022.

[7] Zhutian Yang, Caelan R. Garrett, Tomás Lozano-Pérez, Leslie Kaelbling, and Dieter Fox. Sequence-based plan feasibility prediction for efficient task and motion planning. In *Robotics: Science and Systems (RSS)*, 2023.